

## PROJECT DOCUMENTATION

# Counselor NIDS

A network intrusion detection system in which machine-learning detectors that advise each other and the Snort rule engine watch the same live traffic, and in which Snort can guide the detectors on a network they were never trained on.

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Final-year thesis project · October 2026

Source code: [github.com/hash-cmd/counselor-nids](https://github.com/hash-cmd/counselor-nids)

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# 1. Summary

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Counselor NIDS watches the traffic on a network interface and raises an alarm when a connection looks like an attack. Two independent checks judge every connection:

- **Three machine-learning detectors**, each a specialist in one family of attacks: floods (DoS), break-ins (password guessing and website attacks) and botnets. When a detector is unsure, it asks the others for advice. This is the *counselors network* of Quincozes et al. (2019).
- **Snort 3**, a widely used signature engine that recognises known attack patterns inside the traffic.

Every Snort alert is linked to the connection it belongs to and compared with the AI's verdict on that connection, so each incident is reported as *Confirmed* (both agree), *Disputed* (Snort only) or *AI only*. A web dashboard shows the incidents, who is attacking, how the detectors reached their decisions, how many false alarms the system raises on the network over days, and whether every part of the system is running.

On real attack traffic it never trained on, the system catches 99–100% of most attack types, including floods, password guessing and botnet traffic, with almost no false alarms (Section 8). The thesis extension (Section 9) adds Snort as a counselor, self-learning from agreement between Snort and the AI, and adaptive trust, so the detectors can adapt to a network they were never trained on without anyone labelling its traffic.

# 2. The problem and the thesis objectives

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Machine-learning intrusion detectors score almost perfectly on the dataset they were trained on, but lose most of that accuracy on a different network. Trained on CIC-IDS2017 and tested on CSE-CIC-IDS2018, published classifiers fall to about chance accuracy (Cantone et al., 2024), or a macro-F1 near 0.12. Every real deployment is a "different network", and labelling its traffic by hand is too expensive, so this is the main reason AI-based detection is not trusted in practice.

## GENERAL OBJECTIVE

To design, implement and evaluate a counselor-based network intrusion detection system in which a signature-based IDS (Snort) guides machine-learning detectors to adapt to a network they were never trained on, without labelled traffic from that network.

| Specific objective                 | Challenge it addresses                                                                                                                                                                                     |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Snort as a counselor</b>     | In the counselors network, detectors only ask other AI detectors, who can share the same blind spots on a new network. Snort's signatures behave the same on any network, so they give independent advice. |
| <b>2. Learning from agreement</b>  | There are no labels on the new network. Connections where Snort and the AI independently agree serve as labels for free, and a label-free safety gate blocks harmful retraining.                           |
| <b>3. Adaptive trust</b>           | Each counselor's trust is measured once, in the lab. It is re-estimated on the new network, per detector cluster and per Snort rule, from how often the others agree with it.                              |
| <b>4. Cross-network evaluation</b> | The mechanisms are evaluated by training on one lab (CSE-CIC-IDS2018) and deploying on another (CICIDS2017), in both directions, against each baseline.                                                    |
| <b>5. Live deployment</b>          | Research detectors rarely run on live traffic. This one does, and the practical obstacles found along the way are documented.                                                                              |

### 3. System design

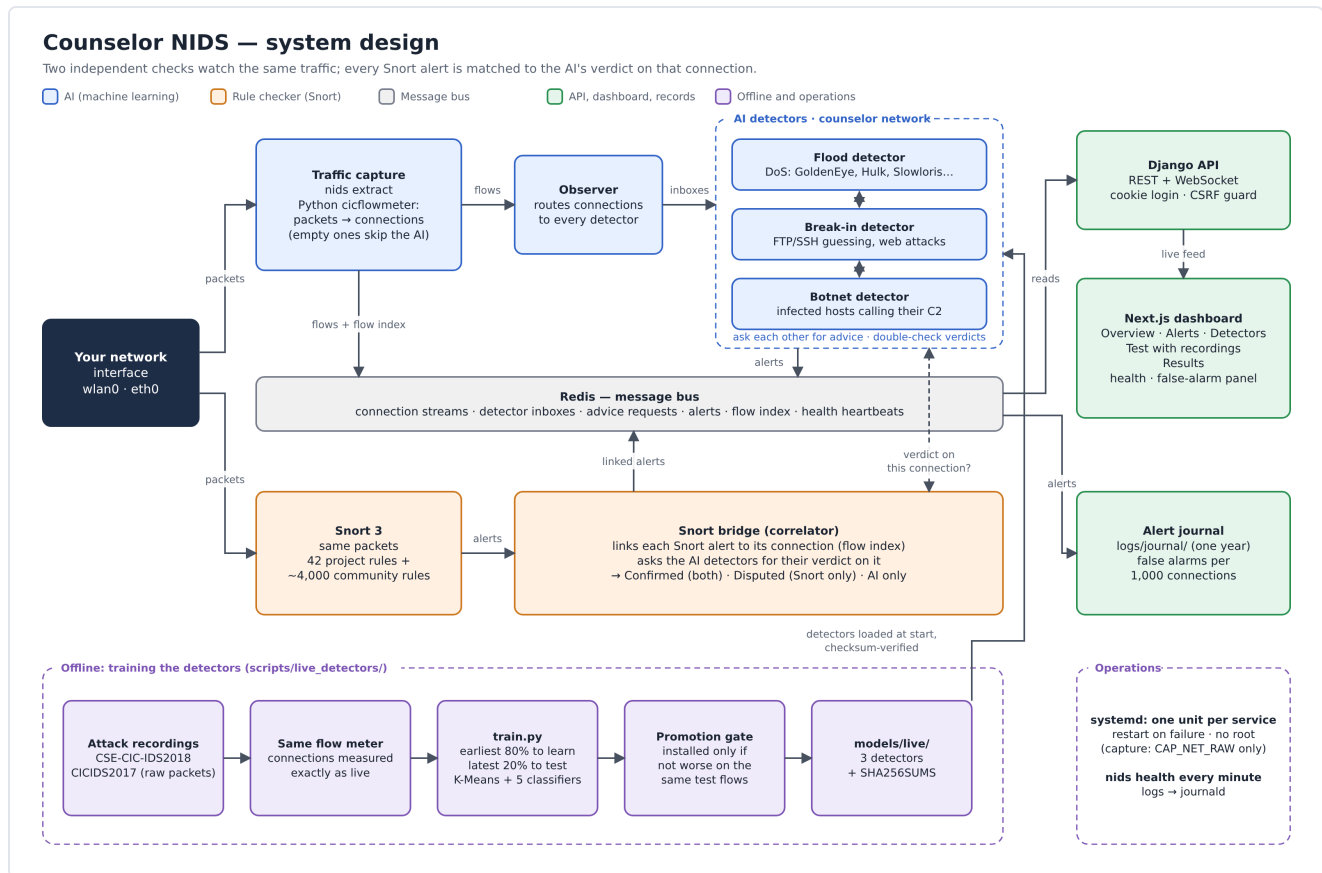


Figure 1. System design: two lanes judge the same packets; Redis carries the messages between services.

| Service                     | Command                                   | Role                                                                                                                                                                                                             |
|-----------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Traffic capture (Extractor) | <code>nids extract -live IFACE</code>     | Captures packets and turns them into connection records (flows) with the Python <code>cicflowmeter</code> . Flows that carry no data go to Snort only. Survives the interface going down and resumes on its own. |
| Observer                    | <code>nids observe</code>                 | Routes each batch of flows to every detector subscribed to that traffic source.                                                                                                                                  |
| Detector (×3)               | <code>nids detect MODEL</code>            | Classifies flows, asks the other detectors for advice and answers their requests, and publishes alerts.                                                                                                          |
| Snort bridge                | <code>nids snort --interface IFACE</code> | Runs Snort on the same interface, links each alert to its connection and asks the detectors for their verdict on it.                                                                                             |
| Alert journal               | <code>nids journal</code>                 | Keeps every alert on disk (one file per day) to measure false alarms over days.                                                                                                                                  |
| Health                      | <code>nids health</code>                  | Every service sends a heartbeat; this lists them and exits with an error if any part has stopped.                                                                                                                |
| API and dashboard           | Django (Daphne), Next.js                  | REST and WebSocket API and the web dashboard.                                                                                                                                                                    |

#### TECHNICAL NOTE

All services communicate through Redis streams: `nids:unknown` (new flows), `nids:inbox:<detector>`, advice request and reply lists, `nids:alerts` and `nids:snort:alerts`, a flow index (sorted sets keyed by connection and by host pair), and a `nids:health` hash of heartbeats. Because each part is a separate process, any one can crash and restart without stopping the others.

## 4. The AI detectors and the counselors network

### How one detector decides (from the paper)

- Training (Algorithm 1).** Five classifiers are trained: naive Bayes, three kinds of decision tree (J48-like, REPTree-like, RandomTree-like) and a random forest. The evaluation data is grouped into clusters with K-Means. For every cluster, the most accurate classifier, and any within 0.1% of it, is kept.
- Detection (Algorithm 2).** A new connection is placed in its nearest cluster, and that cluster's selected classifiers vote. If they agree, that is the verdict. If they disagree, there is a *conflict*.
- Counselors.** On a conflict, the detector asks the other detectors (its counselors) for their verdict on the same connection. It takes the advice of the most confident counselor, if that counselor's accuracy is at least 90%.

## Extensions made in this project

- **Cross-check.** A detector cannot raise a conflict about an attack family it never learned; it is just confidently wrong. So every "normal" verdict is also checked with the counselors, and a confident "attack" advice overrides it.
- **Unresolved conflicts count as normal.** When the classifiers disagree and no counselor can help, the paper falls back on the best classifier's guess. On live traffic those guesses caused most false alarms while accounting for almost no real detections, so they are treated as normal.
- **Connection-only flows skip the AI.** Port scans and failed connections exchange no data. The detectors never learned them and only produced guesses (one scan caused ~50,000 false alarms), so they are left to Snort.

| Detector          | Model file                                  | Learned                                                                             |
|-------------------|---------------------------------------------|-------------------------------------------------------------------------------------|
| Flood detector    | <code>models/live/live_dos.joblib</code>    | DoS GoldenEye, Hulk, Slowloris, SlowHTTPTest (and DDoS once the 2017 data is added) |
| Break-in detector | <code>models/live/live_access.joblib</code> | FTP and SSH password guessing, website attacks                                      |
| Botnet detector   | <code>models/live/live_bot.joblib</code>    | Ares-infected hosts calling their command-and-control server                        |

## 5. Snort alongside the AI

Snort 3 runs on the same interface with the project's 42 rules ( `snort/rules/nids.rules` ) and the ~4,000 Snort 3 community rules. Each alert is linked to its connection: the same two addresses and ports, the same protocol, and a time inside the connection's start-to-end window. A port-scan alert stands for many probes, so it is linked to every connection between the two hosts within  $\pm 30$  seconds. The detectors are then asked for their verdict on that connection, using the same advice request they use with each other.

| Agreement        | Meaning                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Confirmed</b> | Snort and the AI both say attack: very likely real.                                                                                                                                  |
| <b>Disputed</b>  | Snort says attack, the AI says normal: a possible Snort false alarm, or an attack the AI never learned (for example SQL injection, which looks like ordinary web traffic to the AI). |
| <b>AI only</b>   | The AI flagged a connection Snort had no alert for: typically floods and botnet traffic.                                                                                             |

## Rules and tuning

- **Website attacks:** SQL injection (UNION SELECT; a quote followed by an AND/OR comparison; ORDER BY probing; `information_schema` ), cross-site scripting ( `<script` , event handlers, `javascript:` ) in the URL or a form, and website login brute force (20 password posts from one source in 60 s).

- **FTP and SSH brute force** over a wide 120 s window, so slow guessing paced to slip under a short-window threshold is still caught. **SYN and HTTP floods** are covered by rate thresholds, **port scans** by Snort's scan inspector.
- **Checksums are not checked.** Network cards fill checksums in themselves (offloading), so a machine's own outgoing packets look broken to Snort, which used to drop them. Until this was fixed, Snort raised no website-attack alert at all on the 2018 website-attack capture.
- **Quieted rules:** protocol-anomaly alerts are off, and informational community rules (pings, ICMP errors, 1-minute DNS TTLs, UPnP discovery) are suppressed. They fired ~3,800 times on a normal workstation.



## 6. What happens to each connection

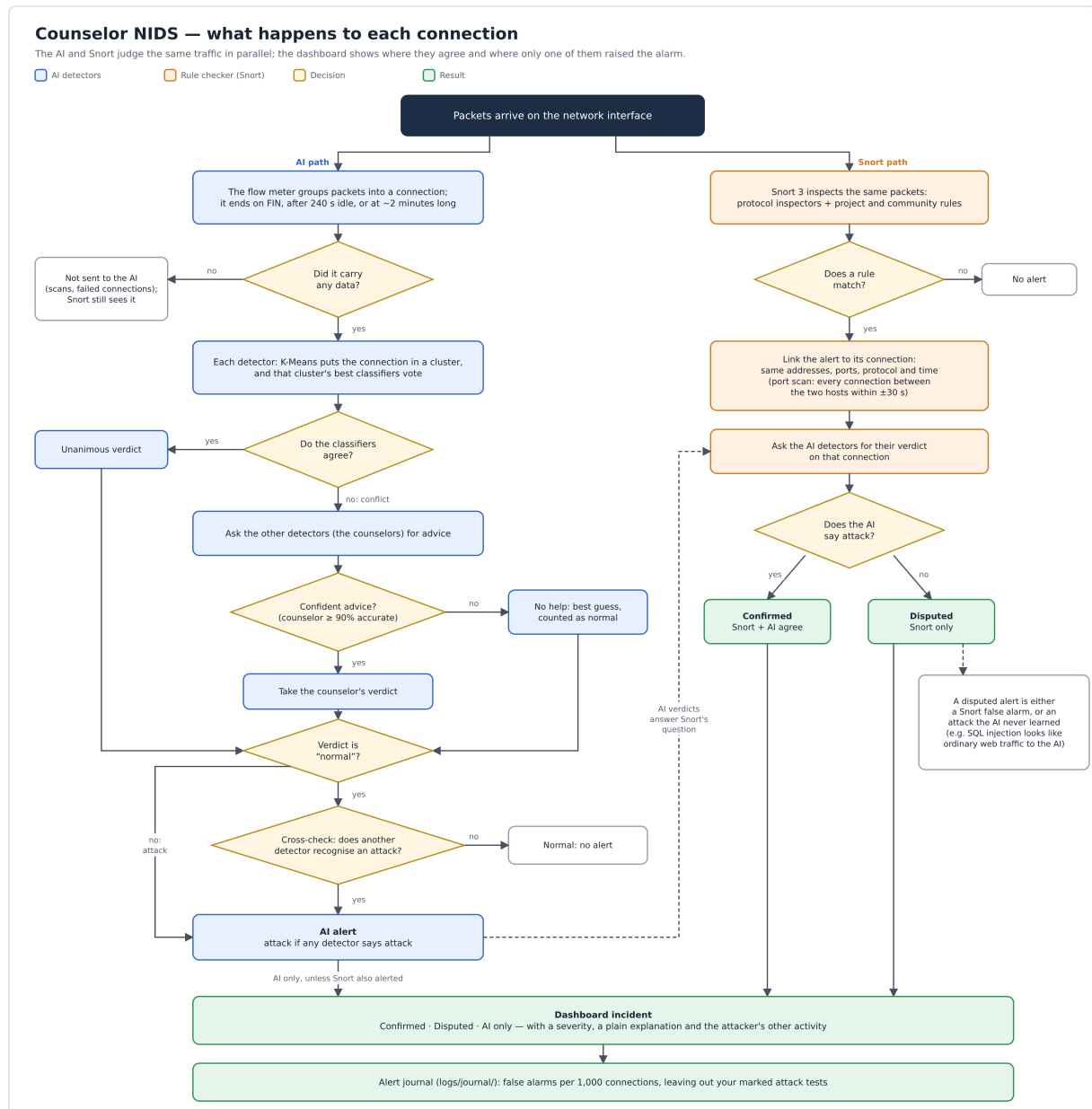


Figure 2. The decision path of one connection through the AI (left) and Snort (right).

## 7. Training the detectors

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### KEY FINDING

Detectors trained on the published dataset CSVs caught **0%** of password-guessing attacks measured the live way. The CSVs were made with the Java CICFlowMeter; live capture uses the Python cicflowmeter, and the two differ in 71 of 154 feature medians (packet lengths with or without headers, flag counting, where flows are split). So the detectors are trained on the datasets' **raw packet recordings**, measured by the same flow meter as live capture.

1. **Fetch:** single hosts' captures from the CSE-CIC-IDS2018 archive (about 1.3 GB, using HTTP range requests, out of 40–60 GB archives).
2. **Label:** the attack windows are found in the packets themselves, and each flow is labelled by its attacker-victim pair. The dataset's CSV timestamps can't be used: they're a 12-hour clock without AM/PM.
3. **Measure:** the Python flow meter, reading recordings by their own clock (Section 12 explains the bug this fixed). Snort also runs on every slice, and the rules that fired are kept with each flow.
4. **Split by time:** within every attack type and capture, the earliest 80% of flows train and the latest 20% test, so test flows come from later in the attack than anything learned.
5. **Promotion gate:** the installed detectors are tested on the same test flows. New ones replace them only if false alarms don't rise and no attack type drops by more than 2 points. Model files are pickles, which run code when loaded, so the installed set is listed with SHA-256 checksums that live mode verifies before loading.

## 8. Results

### Test split: the latest 20% of every attack (CSE-CIC-IDS2018)

| Traffic                              | Test flows   | Detected by the system |
|--------------------------------------|--------------|------------------------|
| DoS SlowHTTPTest                     | 21,110       | 100.00%                |
| DoS Hulk                             | 5,440        | 99.94%                 |
| FTP brute force                      | 4,416        | 100.00%                |
| Botnet                               | 3,432        | 99.83%                 |
| DoS GoldenEye                        | 3,405        | 99.88%                 |
| DoS Slowloris                        | 2,166        | 84.53%                 |
| SSH brute force                      | 1,015        | 99.01%                 |
| Website attacks                      | 42           | 92.86%                 |
| <b>Normal traffic (false alarms)</b> | <b>5,192</b> | <b>0.04% (2 flows)</b> |
| <b>All attacks</b>                   |              | <b>99.12%</b>          |

### Whole system on a held-out recording

`real-attacks-2018.pcap` holds the final minutes of every attack, which the detectors never trained on, plus normal office traffic. It runs through the flow meter, the detectors, Snort with the project's rules, and the alert linking.

| Traffic                      | Flows        | AI          | Snort       | Either      |
|------------------------------|--------------|-------------|-------------|-------------|
| DoS GoldenEye                | 1,822        | 99.1%       | 4.9%        | 99.1%       |
| DoS Hulk                     | 2,861        | 99.9%       | 48.2%       | 99.9%       |
| DoS SlowHTTPTest             | 4,450        | 100.0%      | 99.8%       | 100.0%      |
| DoS Slowloris                | 760          | 34.7%       | 0.0%        | 34.7%       |
| FTP brute force              | 5,040        | 100.0%      | 99.8%       | 100.0%      |
| SSH brute force              | 359          | 98.3%       | 88.9%       | 98.9%       |
| Website attacks              | 4            | 4 of 4      | 4 of 4      | 4 of 4      |
| Botnet                       | 540          | 99.6%       | 0.0%        | 99.6%       |
| <b>Normal (false alarms)</b> | <b>1,570</b> | <b>0.0%</b> | <b>0.3%</b> | <b>0.3%</b> |

- **The two checks cover each other.** The AI catches floods and the botnet that Snort's rules miss. Snort covers website attacks: on the whole 2018 website-attack capture it alerts on 188 of the attacker's 207 connections (90.8%) and on nothing else.

- **A botnet host it never saw:** 99.8% of 15,238 botnet flows caught, with false alarms on 2 of 4,307 normal flows.
- **Slowloris is the weak spot,** and the rate depends on the attack's phase: 85% across the attack, 60% over its complete last five minutes, 35% in the last three as the attack winds down. Snort's Challenge-ACK community rule lifts the last three minutes to 69.5%.

## 9. The thesis extension: Snort-guided adaptation

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### 9.1 Snort as a counselor

The counselors network now accepts *advisors*: counselors that give advice but classify nothing themselves. Snort is one. When a detector's classifiers disagree on a connection, or it cross-checks a "normal" verdict, Snort's verdict on the same connection is advice like any other. Snort only ever says "attack", so it advises only on connections it alerted on; silence is not evidence of normal traffic, since Snort cannot see floods or botnets. The advice is worth the trust of the most trusted rule that fired.

### 9.2 Learning from agreement

While the detectors watch a new network, each chunk of traffic produces *agreement labels*:

- **attack:** Snort alerted and the AI says attack *without Snort's advice* (so Snort cannot confirm itself);
- **normal:** Snort is silent and every detector is unanimously sure the connection is normal.

Everything else stays unlabelled. Detectors retrain on these labels; a fifth of them is held back, and a retrained detector replaces the old one only if it is at least as accurate on the held-back labels. This label-free gate uses no ground truth.

### 9.3 Adaptive trust

- **Detectors:** each cluster's classifier accuracies, which set both the detector's advice confidence and its choice of classifiers, are re-estimated on the new network from agreement labels. The lab estimate acts as a prior worth 200 samples.
- **Snort rules:** each rule's trust is a Beta estimate (prior 0.9, worth 20 observations), raised when the AI independently agrees and lowered when every detector is sure the connection is normal. A rule the AI keeps contradicting, like the DNS-TTL rule on normal lookups, quickly loses its say.

### 9.4 The experiment

`scripts/live_detectors/adapt.py` trains the detectors on one lab and streams the other lab's traffic through the system in time order, in chunks. A fixed test set (the target's latest 20%) is scored before adaptation and after every chunk. It compares:

| Configuration             | Uses                                          |
|---------------------------|-----------------------------------------------|
| Snort alone · AI or Snort | Baselines                                     |
| ai                        | The counselors network as deployed today      |
| +snort                    | + Snort as a counselor                        |
| +trust                    | + Snort counselor and adaptive trust          |
| +learn                    | + Snort counselor and learning from agreement |
| full                      | All three                                     |

It reports detection rate, false-alarm rate, per-attack detection and a balanced score at every step; how many agreement labels each chunk produced and how many were right; and the final trust of every Snort rule.

#### STATUS

All three mechanisms and the experiment harness are implemented. The cross-network results need the CICIDS2017 packet captures (about 50 GB, being downloaded); the builder for that data also runs Snort on every day automatically. Results will be added when the runs complete.

## 10. The dashboard and the API

| Page                 | What it shows                                                                                                                                                                                                                               |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Overview             | What is being watched, and the health of every service; headline numbers with a plain-language verdict; top-risk attackers; AI and Snort detections over time; false alarms on your network, with attack-test marking; the latest incidents |
| Alerts               | Every flagged connection once, with a severity and a plain explanation. An Attackers view groups alerts by source into ranked attack stories, with full-history search and a Print/PDF report                                               |
| Detectors            | Each detector's counters and how its decisions were made: on its own, by advice, double-checked or a best guess                                                                                                                             |
| Test with recordings | Replay a recording through the AI and Snort, apart from live monitoring                                                                                                                                                                     |
| Results              | The detectors' test results per attack type                                                                                                                                                                                                 |

- Written for non-experts ("AI" and "rule checker" rather than "ML" and "Snort").
- A notification bell for new high-severity alerts.
- **Block this IP** writes out a ready-to-paste firewall command; it never runs anything itself.
- Offline IP reputation from public blocklists, looked up locally, so no address leaves the machine.

The API offers login with httpOnly cookies (plus JWT tokens for scripts), detector counters and service health, alerts and incidents, Snort comparisons, the false-alarm journal and attack-test marking, results, IP reputation, replay control, and a WebSocket live feed that sends stats, alerts and health every second.

## 11. Deployment, operations and security

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- **One command:** `./start.sh setup`, then `./start.sh live wlan0`, or `./start.sh --prod live wlan0` for production servers.
- **As services:** `sudo deploy/install-systemd.sh wlan0` installs one systemd unit per part. Each restarts on failure, and the detectors restart with capture so two runs never mix. Nothing runs as root: capture and Snort get only `CAP_NET_RAW` and `CAP_NET_ADMIN`.
- **Health:** every service sends a heartbeat. The dashboard shows one chip per part, and `nids health` runs every minute from a timer.
- **Logs:** the system journal under systemd. Under `start.sh`, logs over 20 MB are rotated. The alert journal keeps one year.
- **Security:** httpOnly SameSite=Strict session cookies; a custom header on every state-changing request as a CSRF guard; at most 10 login attempts a minute per client. In production the API refuses to start without a real secret key, and it sends security headers. Model files are checksum-verified before loading.

## 12. Limitations

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- The detectors learned from one lab with one attacker per attack type. This is exactly the problem the thesis extension tackles; until it is evaluated, check false alarms on your own network with the journal.
- The botnet detector knows one botnet family (Ares). There is no live DDoS or infiltration detector yet; DDoS comes with the 2017 data.
- Slowloris detection varies with the attack's phase (35–85%).
- Website attacks rely mainly on Snort: the AI had only about 200 website-attack flows to learn from.
- Snort-guided adaptation can only use Snort's help for attacks Snort can see. For floods and botnets it depends on the AI detectors agreeing with each other.

## 13. Project layout and how to run it

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```
nids/
├─ start.sh           set up and run everything
├─ deploy/           systemd units and their installer
├─ docs/             deployment, live testing, datasets, diagrams, these PDFs
├─ models/live/      the three detectors, manifest.json, SHA256SUMS
├─ snort/            Snort 3 configuration and the project's rules
├─ src/nids/         capture, detector, counselor (incl. Snort counselor), services,
cli
├─ scripts/live_detectors/ fetch, build, train, evaluate, adapt (the thesis experiment)
├─ backend/          Django API: REST, WebSocket live feed, replays, login
└─ frontend/         Next.js dashboard
```

```
./start.sh setup          # once
./start.sh live wlan0     # watch the network, dashboard on http://localhost:3000
.venv/bin/nids health     # are all parts running?
.venv/bin/nids journal-report # false alarms over days
python scripts/live_detectors/adapt.py --source 2018 --target 2017 # thesis experiment
```

## 14. References

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- M. Cantone et al., "On the Cross-Dataset Generalization of Machine Learning for Network Intrusion Detection", 2024 (arXiv:2402.10974).
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- Snort 3 and the Snort 3 community rules, Cisco Talos (snort.org).